Name: Test Case - OCR spelling errors Class: 10-B Subject: Combined Paper Total: 15 marks

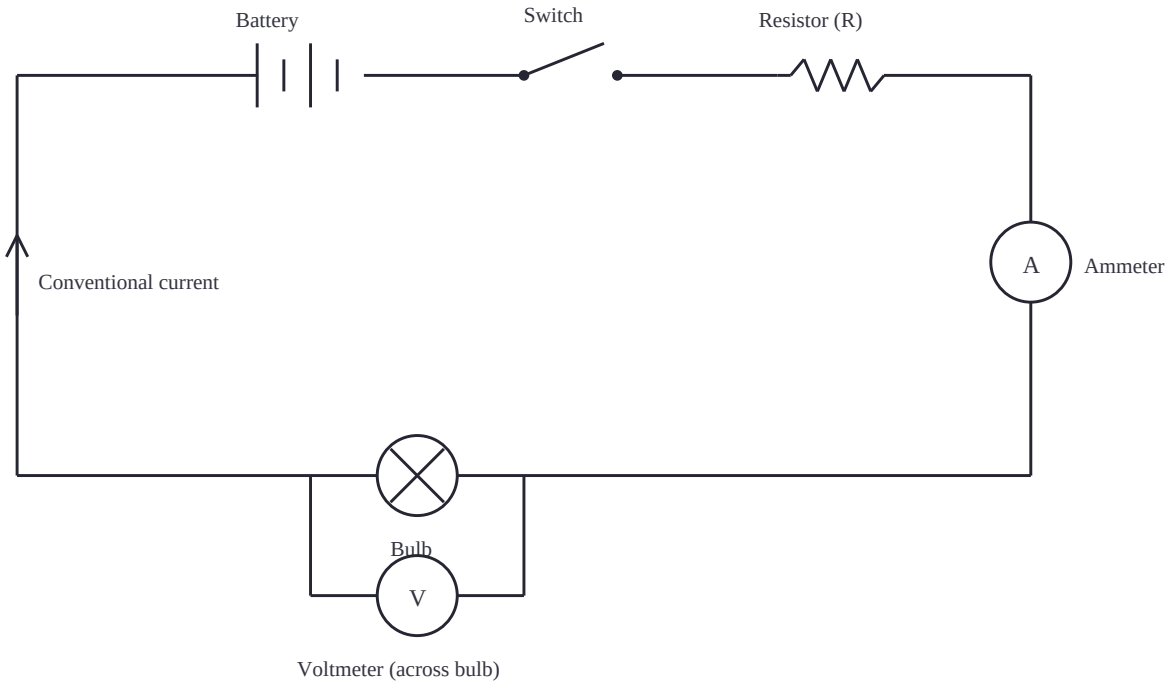
Question 1

A simple electric circut provides a closed path through which electric curent can flow. The battery provides the potentail difference that drives the curent around the circut. The switch opens or closes the circut: when it is closed there is a complete conducting path and curent flows through the bulb and the resistor, and when it is open the path is broken and no curent flows.

The battery, switch, resistor, bulb and amether are all connected in series in the main circut. The amether is connected in series becuase it measures the curent flowing through the circut. The voltmeter is connected in parallel across the bulb, becuase it measures the potentail difference between the two ends of the bulb.

The amount of curent depends on the potentail difference and on the resistence. By Ohm's law, V = IR. If the voltage of the battery stays constant and the resistence is increased, the curent flowing through the circut decreases; reducing the resistence allows more curent to flow.

The diagram below is labelled with every component and shows the direction of conventional curent from the positive terminal of the battery through the external circut back to the negative terminal.



Question 2

In my opinion technology does not automatically make studnets better or worse learners; its effect depends entirely on how it is used. Information that was once difficult to reach is now available in seconds, and a student who cannot follow a topic in class can watch several explanations until one of them makes sense. For example, a student struggling with the water cycle can compare three short videos and find the explanation that finally fits.

However, easy access also creates a real danger. Students may become dependent on searching for an answer instead of working a problem out for themselves, and a student who looks up every difficult question may finish the homework without ever developing the reasoning that the homework was set to build. There is also the risk of accepting inaccurate information simply becuase it appears online.

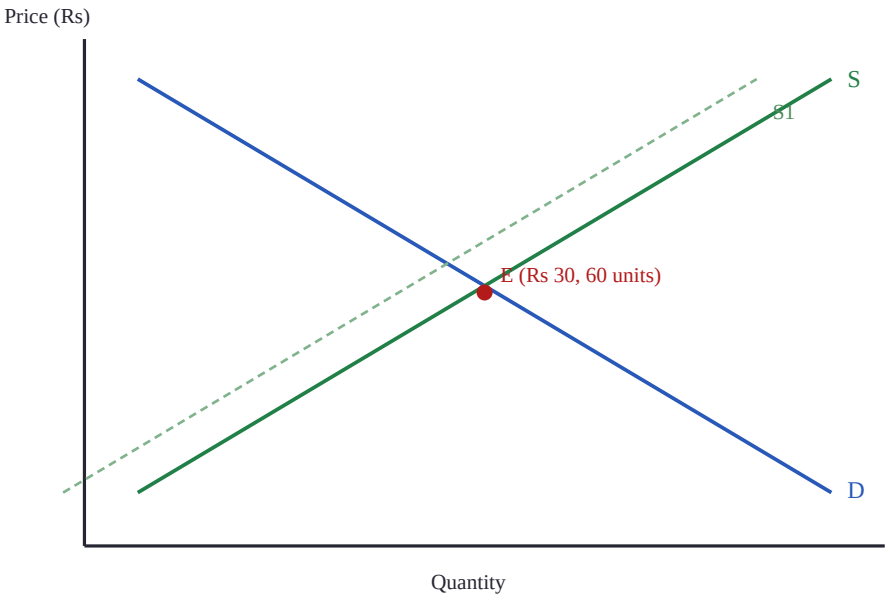
In conclusion, technology is most useful when it is used to understand concepts, explore ideas and check one's

own work, rather than to copy answers. It should therefore be treated as a tool that supports thinking, not as a replacement for thinking.

Question 3

I have plotted the schedule with the quantitiy on the horizontal axis and the price on the vertical axis. The demand curve slopes downward from left to right becuase consumers demand a greater quantitiy at a lower price, and the suply curve slopes upward from left to right becuase producers are willing to suply more at a higher price.

The two curves intersect at a price of Rs 30 and a quantitiy of 60 units. This is the market eqilibrium, becuase at this price the quantitiy demanded is exactly equal to the quantitiy supplied.

If the market price is below the eqilibrium price, the quantitiy demanded is greater than the quantitiy supplied, which creates a shortage and puts upward pressure on the price. If the market price is above the eqilibrium price, the quantitiy supplied is greater than the quantitiy demanded, which creates a surplus and puts downward pressure on the price.

If the cost of production increases, producing becomes less profitable at every price, so producers suply less at each price and the suply curve shifts to the left. The new eqilibrium is therefore at a higher price and a lower quantitiy, assuming demand is unchanged.